Research and developments on the eastern Bering Sea walleye pollock stock assessment

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Summary

The work presented below was based on the same data sets used in the 2024 assessment. We broke the work into two parts, first is the discussion paper presented to the Center for Independent Experts (CIE) review panel prepared in May 2025 (link here). FT-NIR data with the traditional data sources (BTS and ATS). The results of these comparisons are presented below. Pre and post CIE review…

The model fits to the data are presented in the tables at the end of the document. The first section steers towards the R library used for processing model results for EBS pollock. The second section presents the comparisons of the FT-NIR data with the traditional data sources. The third section compares results of different software platforms (e.g., SS3, CEA, and AMAK) for the assessment. The fourth section shows results comparing newer model-based estimates from survey data.

Response to 2025 CIE Review of EBS Walleye Pollock Assessment

This section outlines the response to the findings and recommendations of the 2025 Center for Independent Experts (CIE) review of the Eastern Bering Sea (EBS) walleye pollock stock assessment. The final peer-reviewed reports from the three reviewers are provided at this link https://www.st.nmfs.noaa.gov/science-quality-assurance/cie-peer-reviews/cie-review-2025 . These documents summarize their findings and recommendations in detail.

Since the review:

- The operational ADMB-based model has been fully ported to **RTMB**, with verification that outputs match the prior version.
- Initial steps are underway to address **Russian catch** uncertainty through scenario modeling based on available data.

1. Model Platform and Estimation Framework

The full ADMB model has been implemented in RTMB, enabling: - Random walk selectivity and recruitment deviations - Improved error propagation via marginal likelihood - Mohn's rho, retrospective peels, and full simulation integration

Status: Complete
See attached illustration chunk for output comparison (add figures/tables here).

2. Russian Removals and Stock Exchange

Russian catch has not historically been included in the assessment. All three reviewers noted this omission and recommended: - Conducting **sensitivity runs** - Quantifying the **scale and direction of transboundary movement** - Developing **collaborative or joint modeling strategies**

Status: In progress
Ongoing work involves sensitivity runs with assumed Russian removals of 20–45% of the EBS US catch, guided by UCSL (2024) and reviewer recommendations.

3. Process Uncertainty, Selectivity, and Growth

Key issues raised: - Random walk selectivity penalties are currently fixed. Reviewers suggested estimating process variances via marginal likelihood (Anders). - Weight-at-age models could be expanded to include cohort and temperature effects (Yan). - Estimation of time-varying natural mortality should be explored cautiously (Yan).

Status: Partially addressed — EDF and penalty variance exploration will proceed after Russian catch integration.

4. Data Weighting and Compositional Likelihoods

Matt Cieri strongly recommended testing a **Dirichlet-multinomial (DM)** approach via Laplace approximation to better reflect age composition variance.

Status: Under review — current bootstrap-based effective N retained; DM implementation deferred until after RTMB stabilization.

5. Retrospective Performance

Mohn's rho was computed and incorporated into RTMB outputs. Diagnostic plots are available for: - Spawning biomass (SSB) - Recruitment (R) - Fishing mortality (F)

Status: Complete

6. Summary of Reviewer Recommendations and Status

```
tribble(
  ~Theme, ~`Lead Reviewer`, ~Recommendation, ~Status,
  "RTMB transition", "All", "Port model from ADMB to RTMB", "  Completed",
  "Russian catch", "All", "Incorporate or test Russian removals", "  In progress",
  "Selectivity complexity", "Nielsen", "Estimate process variances via marginal likelihood",
  "Natural mortality", "Jiao", "Explore M variation and CEATTLE integration", "  Conceptual",
  "Growth & WAA", "Jiao", "Model cohort and temperature effects on WAA", "  Deferred",
  "Data weighting", "Cieri", "Test Dirichlet-multinomial likelihoods", "  Deferred",
  "Retrospective diagnostics", "Nielsen", "Use Mohn's rho and integrate peels", "  Completed"
) %>%
  gt::gt()
```

Table 1: CIE 2025 Recommendations and Response Summary

Theme	Lead Reviewer	Recommendation	Status
RTMB transition	All	Port model from ADMB to RTMB	Com
Russian catch	All	Incorporate or test Russian removals	In pr
Selectivity complexity	Nielsen	Estimate process variances via marginal likelihood	Plan
Natural mortality	Jiao	Explore M variation and CEATTLE integration	Conc
Growth & WAA	Jiao	Model cohort and temperature effects on WAA	Defe
Data weighting	Cieri	Test Dirichlet-multinomial likelihoods	Defe
Retrospective diagnostics	Nielsen	Use Mohn's rho and integrate peels	Com

ADMB → RTMB Code

subtitle: "Tracking translation details while bridging ADMB to RTMB"

This standalone appendix tracks items to verify or refactor as the model is bridged from **ADMB** to **RTMB**. Each entry records the ADMB behavior, intended RTMB behavior, priority, and links/tests.

Model implementation

The RTMB model code is implemented in the `R/` directory, with the main entry point in `R/Rpm.R`.

Comparison of ADMB and RTMB by components

The initial steps to bridge ADMB to RTMB involve identifying key components and their behaviors. The following table summarizes the components, their ADMB behavior, intended RTMB behavior, and current status (Table 1).

After some initial testing we were able to take the parameter estimates from the ADMB model and show that the gradients were very close to zero and the negative log-likelihood values matched.

outer mgc: 6.026919e-06 outer mgc: 6.026919e-06

Reparameterizing dev-vectors

The ADMB code has lots of cases where say y_i is a function of a parameter θ and a deviation ϵ_i such that $y_i = \theta + \epsilon_i$. In ADMB, the deviations are estimated as a vector of parameters with a constraint to have mean zero., In RTMB, it seems best to reparameterize them as a simple vector of parameters (say as θ_i) and then do the likelihood parts as deviations from the mean.

So far we've done this for the following parameters:

- `log_avgrec` (average recruitment)

Table 1: Comparison of the rtmb code (`RPM`) and the ADMB code results.

Opening /Users/jim/__mymods/noaa-afsc/EBS__pollock/runs/data/wtage2024.dat
[

Comparison of RTMB and ADMB Outputs
Tolerance = 1e-05] Comparison of RTMB and ADMB Outputs
Tolerance = 1e-05

Variable	Equal (tol)	Length	Max |Abs diff|	Max |% diff|	Correlation
N	TRUE	915	0.049666	0.000439	1.000000
Z	TRUE	915	0.000000	0.000151	1.000000
F	TRUE	915	0.000000	0.000491	1.000000
M	TRUE	915	0.000000	0.000000	1.000000
S	TRUE	915	0.000000	0.000120	1.000000
pred__catch	TRUE	61	0.004832	0.000391	1.000000
obs__catch	TRUE	61	0.005000	0.000458	1.000000
SSB	TRUE	61	0.004954	0.000428	1.000000
phizero	TRUE	1	0.000000	0.000189	NA
Bzero	TRUE	1	0.000761	0.000013	NA
steepness	TRUE	1	0.000000	0.000054	NA
pred__cpue	TRUE	12	0.004774	0.000137	1.000000
pred__avo	TRUE	18	0.000005	0.000314	1.000000
eb__bts	TRUE	42	0.022933	0.000212	1.000000
eb__ats	TRUE	19	0.004530	0.000149	1.000000
sel__fsh	TRUE	915	0.000005	0.000467	1.000000
sel__bts	TRUE	645	0.000000	0.000398	1.000000
sel__ats	TRUE	465	0.000005	0.000491	1.000000
sam__fsh	TRUE	60	0.000499	0.000134	1.000000
sam__bts	TRUE	42	0.000000	0.000000	1.000000
sam__ats	TRUE	19	0.000000	0.000000	1.000000
phat__fsh	TRUE	900	0.000000	0.000476	1.000000
phat__bts	TRUE	630	0.000000	0.000468	1.000000
phat__ats	TRUE	285	0.004637	0.000442	1.000000
age__like	TRUE	3	0.001383	0.000212	1.000000
age__like__offset	TRUE	3	0.039496	0.000326	1.000000
cat__like	TRUE	1	0.000002	0.000078	NA
bts__like	TRUE	1	0.000004	0.000011	NA
ats__like	TRUE	1	0.000003	0.000027	NA
ats__age1__like	TRUE	1	0.000039	0.000346	NA
cpue__like	TRUE	1	0.000002	0.000174	NA
avo__like	TRUE	1	0.000002	0.000020	NA
avgsel__like	TRUE	1	0.000000	0.000235	NA
wt__nll	TRUE	4	0.004116	0.000190	1.000000
wt__like	TRUE	1	0.004805	0.000076	NA
rec__like	TRUE	7	0.000047	0.000460	1.000000
Fpen__like	TRUE	1	0.000004	0.000043	NA
sel__like	TRUE	3	0.000023	0.000090	1.000000
sel__like__dev	TRUE	3	0.000338	0.000287	1.000000
Priors	TRUE	4	0.000006	0.000029	1.000000
tot__like	TRUE	1	0.002266	0.000028	NA

Table 2: Top 20 parameters with largest gradients (absolute value) from RTMB using ADMB parameter estimates.

Parameter	gradient
sel_slp_bts_dev	-6.026919e-06
sel_devs_ats	-5.780988e-06
sel_devs_ats	-5.442764e-06
sel_devs_ats	-5.254151e-06
sel_devs_ats	5.210384e-06
sel_devs_ats	5.065771e-06
sel_devs_ats	5.020803e-06
sel_devs_ats	-4.987770e-06
sel_devs_ats	4.963619e-06
sel_devs_ats	4.907284e-06
sel_devs_ats	4.413068e-06
sel_devs_ats	4.319045e-06
sel_devs_ats	-4.250591e-06
sel_devs_ats	4.144523e-06
sel_devs_ats	-4.106040e-06
sel_devs_ats	-3.973683e-06
sel_devs_fsh	-3.950742e-06
sel_devs_ats	-3.928872e-06
sel_devs_ats	3.832048e-06
sel_devs_fsh	3.822259e-06

- `log_avg_F` (average fishing mortality)
- `log_slp_bts` (slope of the selectivity for the bottom trawl)
- `log_a50_bts` (age-50% selected for the bottom trawl)

The results from feeding the MLE parameter estimates from ADMB into the RTMB code gave exactly the same estimates of spawning stock biomass (SSB) and age 1 recruits (Figure 1; Figure 2).

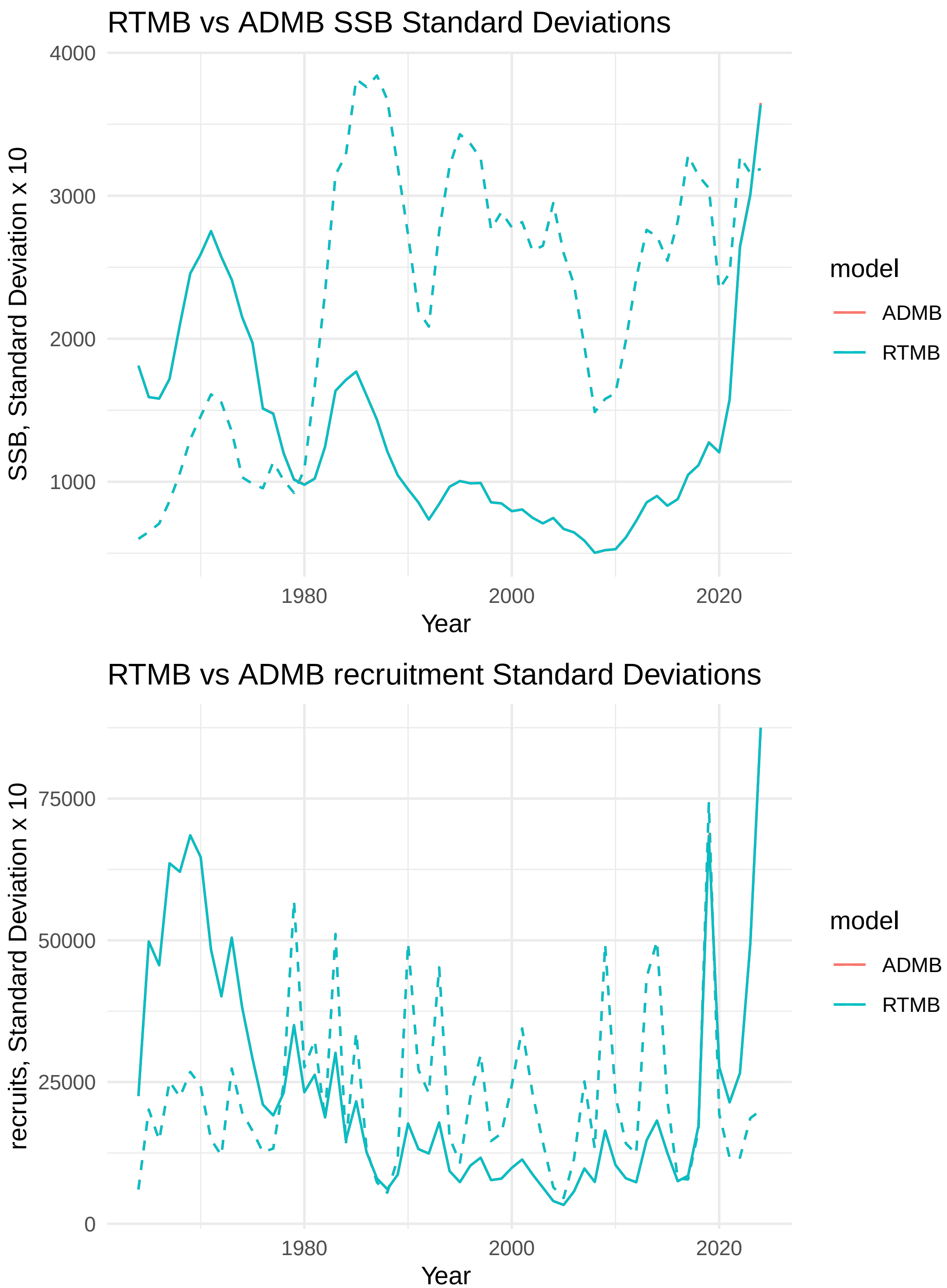

Figure 1: SSB and age-1 recruits from RTMB and ADMB, dashed lines are standard deviations (x10). Both ADMB and RTMB results are overlain.

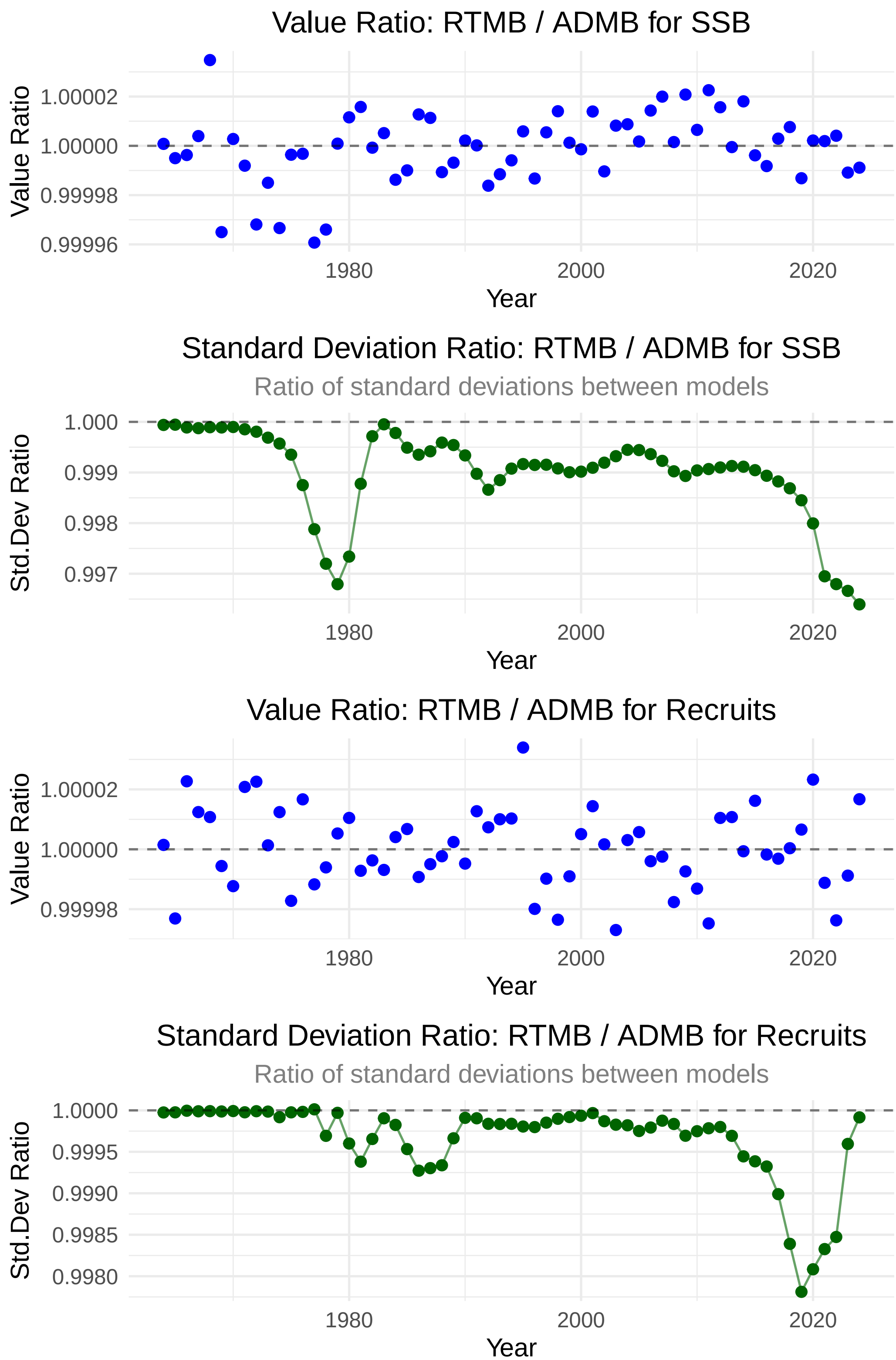

Figure 2: Relative differences between ADMB and RTMB estimates of SSB and age-1 recruits and asymptotic standard deviation approximations.

Considerations for future assessments regarding neighboring Russian federation fisheries

As this year's assessment moves forward, particular attention is warranted on how Russian fisheries and assessments are considered alongside our own results. Pollock in the eastern and western Bering Sea are generally recognized as a single biological stock, with seasonal exchange across the U.S.–Russia boundary documented by acoustic studies. As noted in the CIE reviews, Russian removals are substantial with recent estimates representing on the order of 20–45% of the U.S. catch in some years.
While examined in past assessments, the Russian catch magnitude should be re-evaluated to examine how they might alter management advice. As a step in beginning to make a more detailed evaluation, we compared the Russian stock assessment results presented as part of their Marine Stewardship Council (MSC) review panel with our model results (through 2024). The following sections compare outcomes from the assessments presented to

Comparison of Russian and USA age-structured assessment results

Pollock 99% of catch; seabird & marine mammal mortality very low (Artyukhin, Korobov, and Gluschenko 2023; Burdin 2021). (Japp and Sharov 2023, 2024; Bulatov and Vasilev 2023).

The comparison of U.S. and Russian pollock assessments shows that U.S. estimates of total numbers-at-age are consistently higher than those from Russia, typically by a factor of 2–4, with the ratio peaking above 5 in the early 2010s (Figure 3, bottom-left). Both assessments show similar interannual patterns, particularly for age-1 recruits (top-right), suggesting some shared signals of strong year classes. Correlation analysis by age class (bottom-right; Table 1) confirms this alignment, with moderate to strong positive correlations across ages 1–6. At older ages, correlations weaken and even turn negative (age 10), reflecting differences in how the two assessments treat age structure and fishery/survey data inputs. Overall, while the magnitude of estimates differs substantially between the two countries, the consistent synchrony in recruitment trends indicates that both assessments are capturing the same broad population dynamics, albeit with systematic differences in scale.

Table 1: Correlations between Russian and USA age classes, 1995-2024.

[

Correlations between Russian and USA Age Classes] Correlations between Russian and USA Age Classes

Age	Correlation
1	0.625
2	0.627
3	0.618
4	0.697
5	0.682
6	0.560
7	0.083
8	0.098
9	0.007
10	-0.500

Table 2: Summary statistics of magnitude differences between Russian and USA age classes, 1995-2024.

[1] "mean ratio (USA/Russian): 2.87" [1] "median ratio (USA/Russian): 2.7"

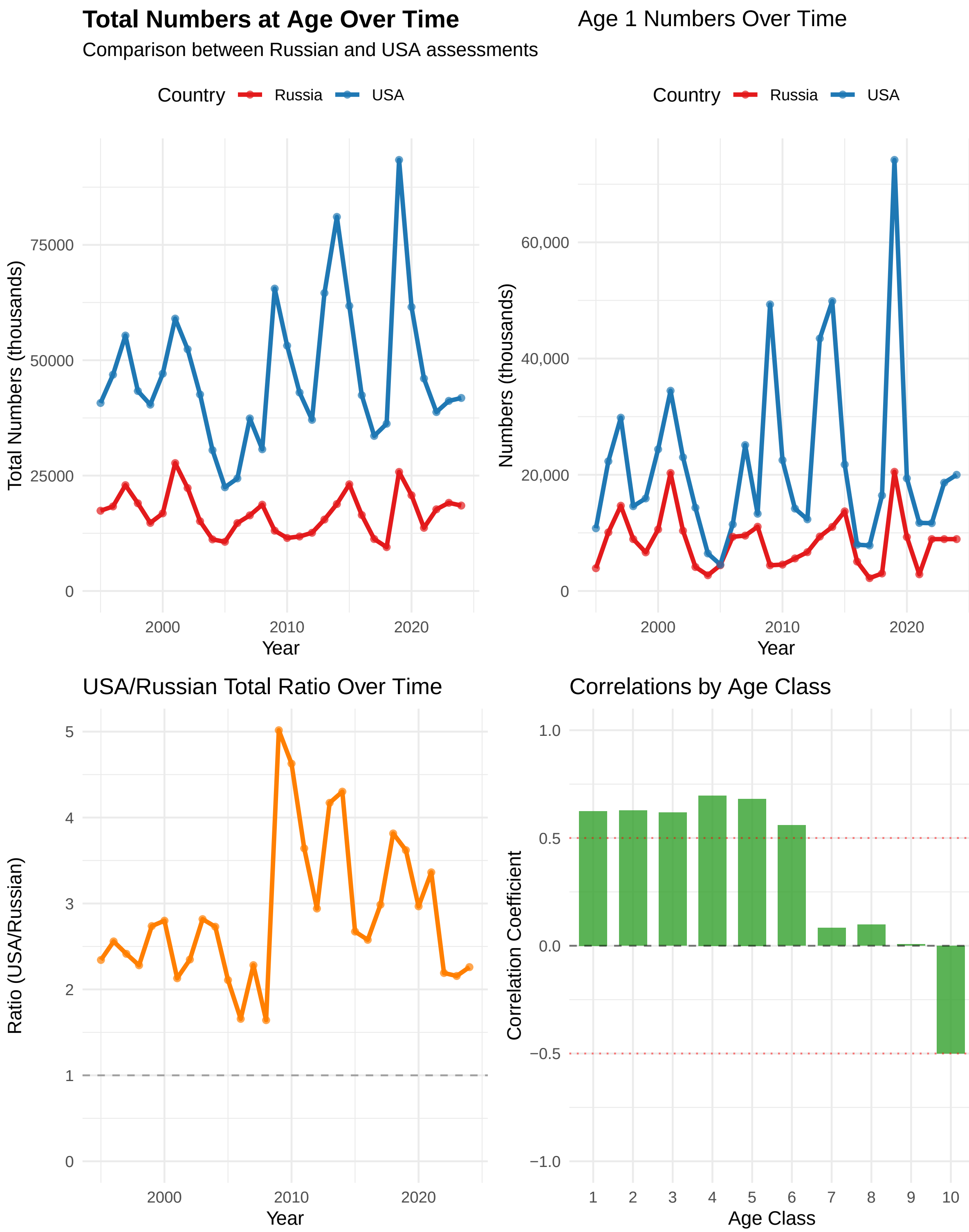

Figure 3: Various comparisons between Russian and USA age-structured survey results.

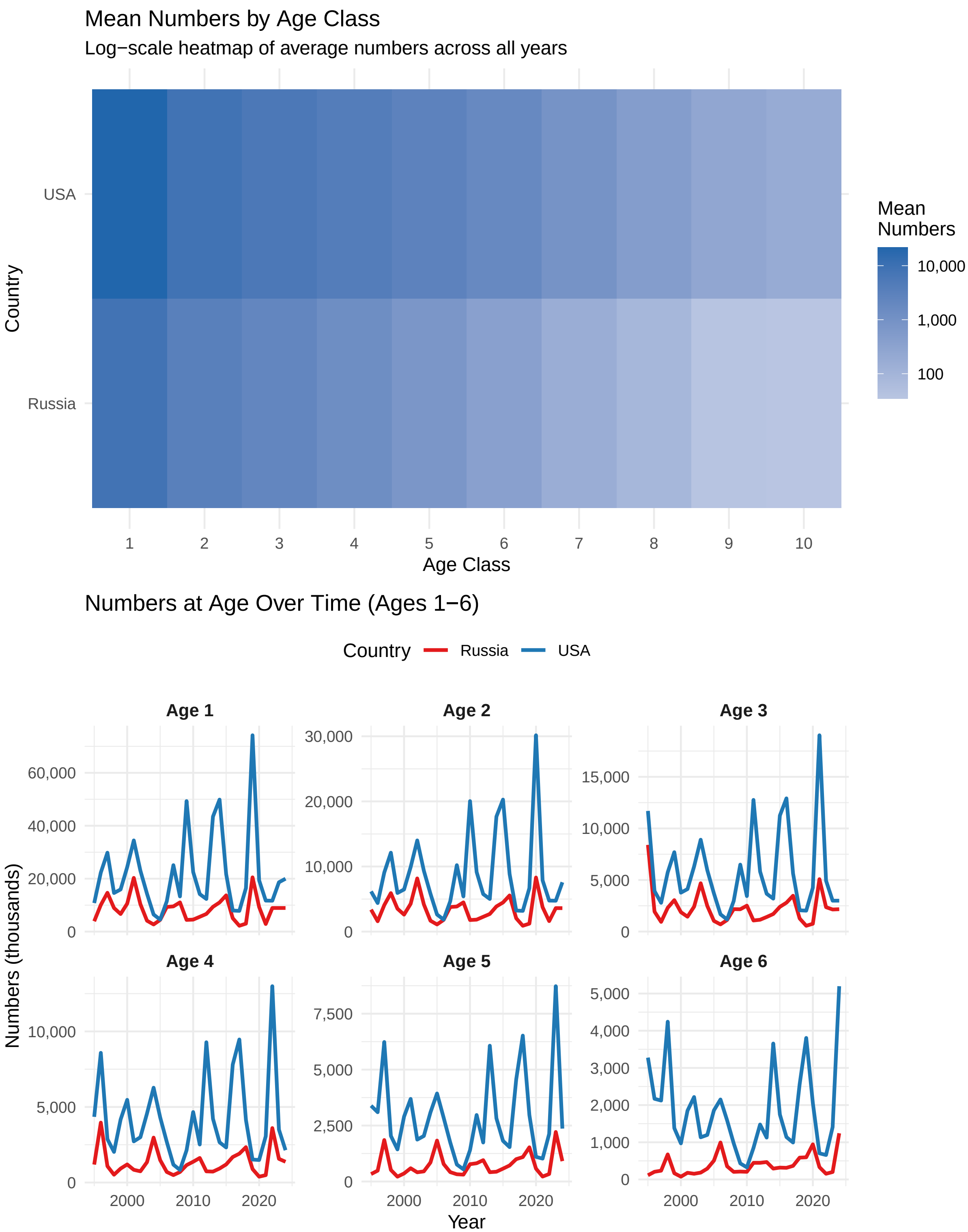

Figure 4: Various comparisons between Russian and USA age-structured survey results.

```
=== DETAILED STATISTICS ===

Summary statistics by country and age:

# A tibble: 20 x 8
   Country   Age    Mean  Median      SD    CV    Min    Max
   <chr>   <dbl>   <dbl>   <dbl>   <dbl> <dbl>  <dbl>  <dbl>
 1 Russia      1  8391.   8925    4647.  0.554 2213.  20509.
 2 Russia      2  3400.   3562.   1860   0.547  900.   8308.
 3 Russia      3  2254.   2154.   1574.  0.699  565.   8406.
 4 Russia      4  1338.   1169.    875   0.654  387.   3966.
 5 Russia      5   743     575.    511.  0.688  216.   2214.
 6 Russia      6   382.    302.    278   0.728   74.5  1242.
 7 Russia      7   168.    138.    127.  0.757   15.3   550.
 8 Russia      8    89.3    65.2    81.9 0.917    0.8   367.
 9 Russia      9    38.7    31.2    37.3 0.964    0.4   178.
10 Russia     10    35.1    16.8    45.4 1.29     0.2   164.
11 USA         1 21725.  17539.  15273.  0.703 4546.  74180.
12 USA         2  8768.   6580.   6227.  0.71  1848.  30158.
13 USA         3  5797.   4186    4098.  0.707 1175.  19030.
14 USA         4  4232.   3283.   2852.  0.674  831.  12994.
15 USA         5  2912.   2594.   1889.  0.648  543.   8729.
16 USA         6  1842.   1535.   1174.  0.637  325    5198.
17 USA         7   935.    809.    565.  0.604  185.   2560.
18 USA         8   488     375.    317.  0.649   88.1  1532
19 USA         9   260.    204.    178.  0.685   41.8   861.
20 USA        10   188     176.    103.  0.547   37.8   476.


Coefficient of Variation by age:

# A tibble: 10 x 3
     Age Russia_CV USA_CV
   <dbl>     <dbl>  <dbl>
 1     1     0.554  0.703
 2     2     0.547  0.71
 3     3     0.699  0.707
 4     4     0.654  0.674
 5     5     0.688  0.648
 6     6     0.728  0.637
 7     7     0.757  0.604
```

```
 8     8      0.917  0.649
 9     9      0.964  0.685
10    10      1.29   0.547


Age classes with correlation > 0.5: Age1, Age2, Age3, Age4, Age5, Age6

Age classes with correlation < 0.2: Age7, Age8, Age9, Age10

  Age1   Age2   Age3   Age4   Age5   Age6   Age7   Age8   Age9  Age10
 0.625  0.627  0.618  0.697  0.682  0.560  0.083  0.098  0.007 -0.500
```

Results comparing Russian and USA age-structured assessments

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Comparison with the Gulf of Alaska (GOA) pollock estimates

Since the Russian context suggest some connectivity, for perspective, we thought it might be useful to conduct a similar comparison with results from the GOA pollock stock. Pollock assessments from the eastern Bering Sea (EBS) and Gulf of Alaska (GOA) show strong contrasts in both magnitude and dynamics of numbers-at-age. The EBS consistently supports far higher abundances across all ages and years (typically 5–15 times greater than the GOA; Figure 5). Recruitment pulses are evident in both regions, particularly at age 1, but the correlations by age class reveal only modest positive associations at younger ages, while older ages show little to no correspondence (Table 3). This indicates that early life stages may share some environmental drivers across basins, but overall population dynamics remain largely decoupled.

A more detailed breakdown by age class underscores these findings (Figure 6). Cohort strength in the EBS exhibits large variability with recurring strong year classes, while the GOA maintains lower overall abundance with weaker pulses. Time-series plots for ages 1–6 highlight this contrast, with the EBS showing sharp recruitment spikes and the GOA remaining more muted.

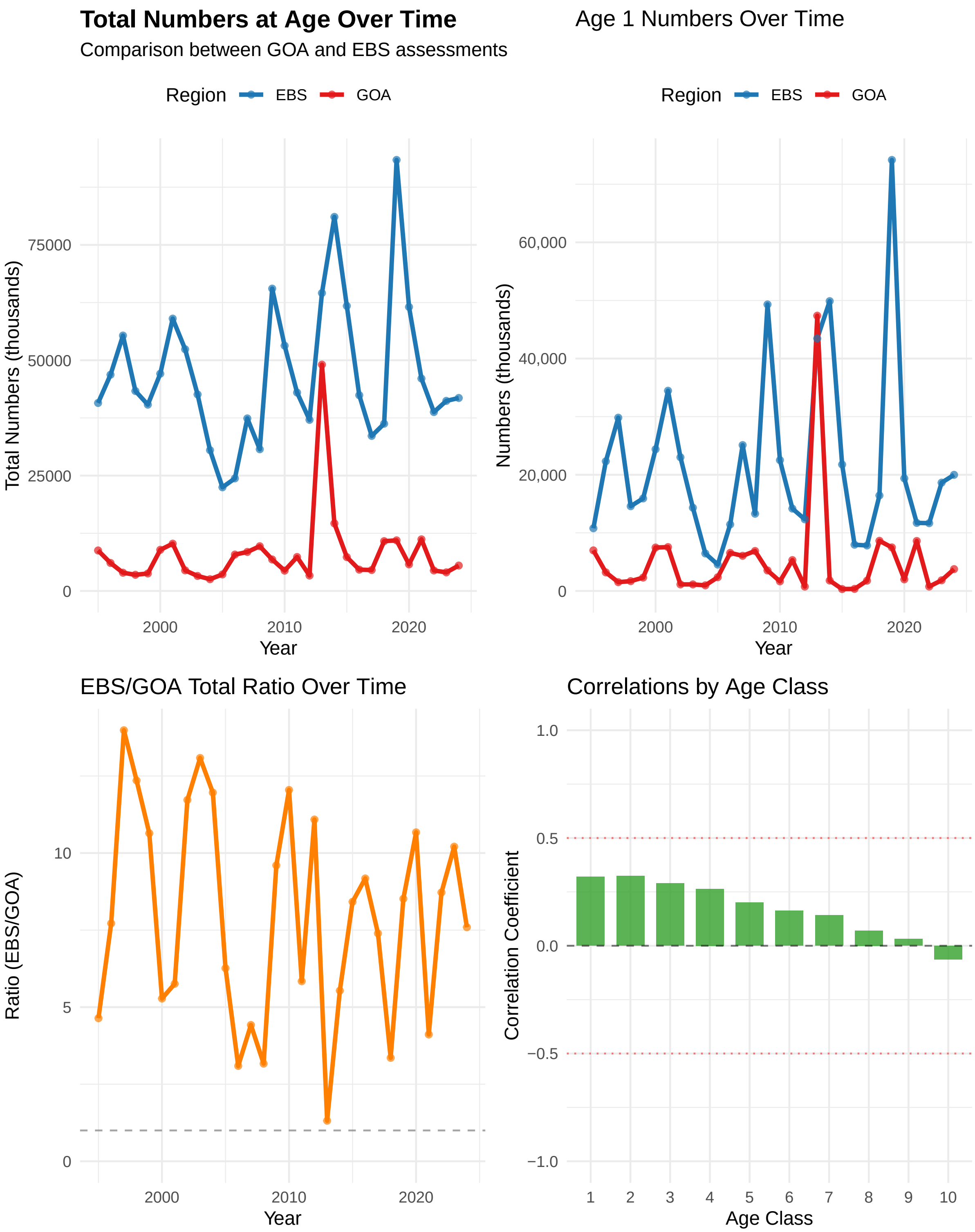

Figure 5: Various comparisons between GOA and EBS age-structured assessment results.

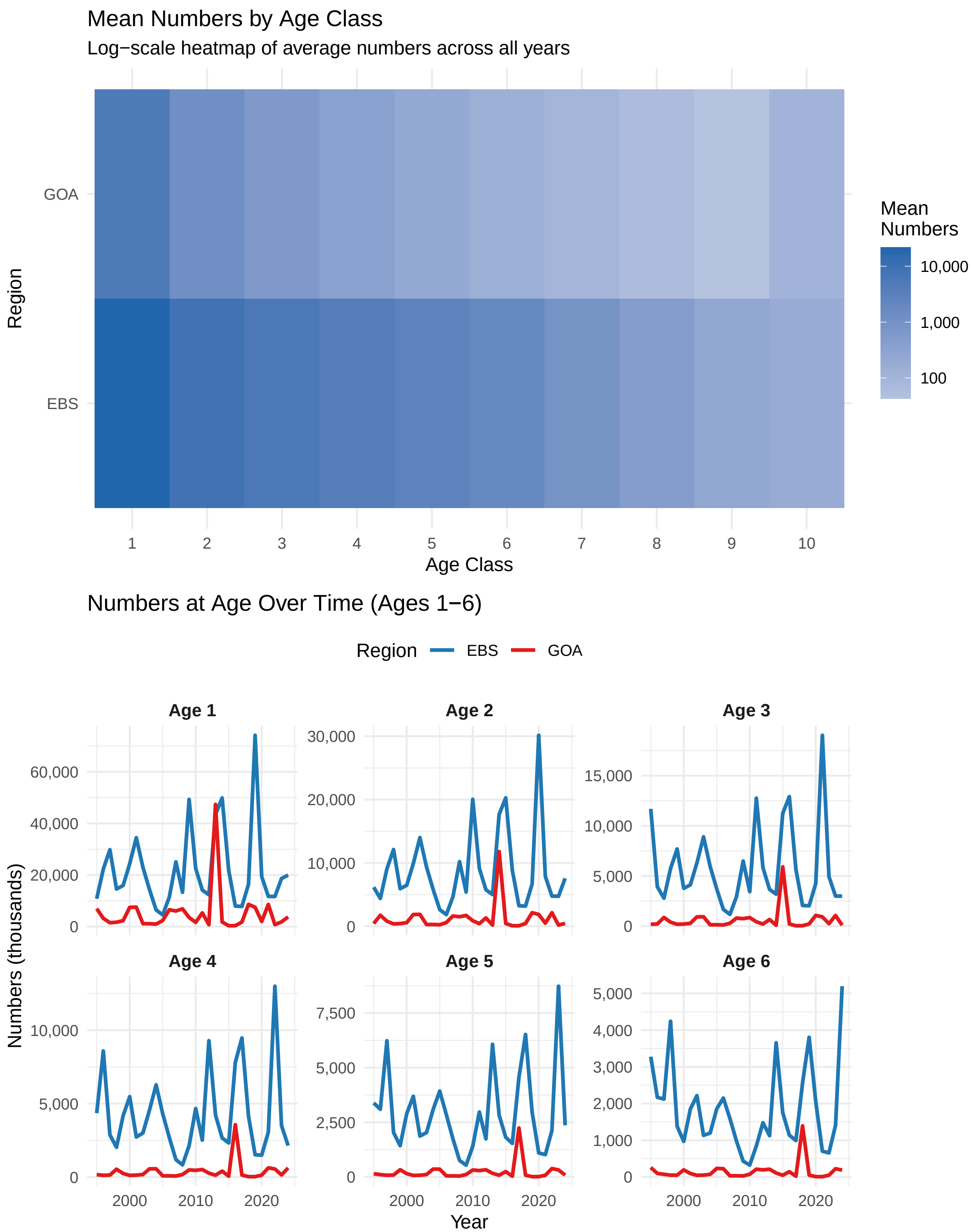

Figure 6: Various comparisons between GOA and EBS age-structured assessment results.

Table 3: Summary of age-specific correlations between GOA and EBS age classes, 1995-2024.

[

Correlations between GOA and EBS Age Classes] Correlations between GOA and EBS Age Classes

Age	Correlation
1	0.321
2	0.324
3	0.289
4	0.264
5	0.200
6	0.162
7	0.142
8	0.071
9	0.032
10	-0.064

Other items

The CIE noted in particular that some documentation of aspects of the ADMB code was incomplete. During the meetings, we began to more properly specify how the selectivity penalties were being calculated and implementated.

First-difference selectivity penalty (temporal)

We need the penalty used in ADMB where the **first year is compared to zero** (not to a previous estimated year). For a matrix of log-selectivities ($\log s_{y,a}$) (years($y = 1, \ldots, T$), ages ($a = 1, \ldots, A$), the penalty is:

$$\mathcal{P}_{\text{FD,time}} = w \sum_{a=1}^{A} \Big[\big(\log s_{1,a} - 0 \big)^2 + \sum_{y=2}^{T} \big(\log s_{y,a} - \log s_{y-1,a} \big)^2 \Big].$$

RTMB/R implementation

```
# log_sel: matrix [T x A] of log-selectivities (year rows, age cols)
# w: scalar or length-A vector of weights
fd_penalty_time <- function(log_sel, w = 1) {
  stopifnot(is.matrix(log_sel))
  T <- nrow(log_sel); A <- ncol(log_sel)
  if (length(w) == 1) w <- rep(w, A)
  stopifnot(length(w) == A)

  diffs <- rbind(log_sel[1, , drop = FALSE] - 0,   # year 1 vs 0
                 apply(log_sel, 2, diff))          # y>=2 diffs
  sum(colSums((sqrt(w) * diffs)^2))
}
```

Sanity check

```
set.seed(42)
T <- 5; A <- 3
L <- matrix(rnorm(T*A, sd=0.2), T, A); w <- c(1, 2, 0.5)

fd_loop <- function(L, w){
  acc <- 0
  for (a in seq_len(ncol(L))) {
    acc <- acc + w[a]*(L[1,a]-0)^2
    for (y in 2:nrow(L)) acc <- acc + w[a]*(L[y,a]-L[y-1,a])^2
  }
  acc
}

all.equal(fd_penalty_time(L, w), fd_loop(L, w))
```

```
[1] "Mean relative difference: 0.01953004"
```

First-difference selectivity penalty (age)

For within-year smoothing over age with a **zero anchor at the minimum age**:

$$\mathcal{P}_{\text{FD,age}} = w \sum_{y=1}^{T} \Big[\big(\log s_{y,a_{\min}} - 0\big)^2 + \sum_{a=a_{\min}+1}^{A} \big(\log s_{y,a} - \log s_{y,a-1}\big)^2 \Big].$$

RTMB/R implementation

```
# log_sel: matrix [T x A] of log-selectivities (year rows, age cols)
# a_min: column index of minimum age within 'log_sel' (default = 1)
fd_penalty_age <- function(log_sel, w = 1, a_min = 1) {
  stopifnot(is.matrix(log_sel))
  T <- nrow(log_sel); A <- ncol(log_sel)
  stopifnot(a_min >= 1, a_min <= A)
  if (length(w) == 1) w <- rep(w, T)
  stopifnot(length(w) == T)

  # Build diffs across age with anchor at a_min (assumes a_min=1; if not, set zeros below a_m
```

```
  diffs <- matrix(0, nrow = T, ncol = A)
  diffs[, a_min] <- log_sel[, a_min] - 0
  if (a_min < A) {
    for (a in (a_min+1):A) {
      diffs[, a] <- log_sel[, a] - log_sel[, a-1]
    }
  }
  sum(rowSums((sqrt(w) * diffs)^2))
}
```

Reference: survey lognormal scaling (consistency check)

When porting lognormal survey likelihoods, confirm whether ADMB's objective includes the normalizing constant and whether ( ) is on the **log** scale:

$\ell(\sigma) = -\sum_i \frac{(\log O_i - \log E_i)^2}{2\sigma^2} \; - \; n \log \sigma \; - \; \frac{n}{2} \log(2\pi).$

```
lognorm_nll <- function(obs, exp, sigma_log, include_const = TRUE) {
  r <- log(pmax(obs, .Machine$double.eps)) - log(pmax(exp, .Machine$double.eps))
  n <- length(r)
  nll <- sum(r^2) / (2 * sigma_log^2)
  if (include_const) nll <- nll + n*log(sigma_log) + n*0.5*log(2*pi)
  nll
}
```

References

Artyukhin, Yuri B., Dmitri V. Korobov, and Yuri N. Gluschenko. 2023. "Incidental Mortality of Seabirds in Pollock Trawl Fishery in the Northwestern Bering Sea." *Trudy VNIRO* 193: 174–89. https://doi.org/10.36038/2307-3497-2023-193-174-189.

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Burdin, A. M. 2021. "Assessment of Impacts of the Pollock Fishery on Steller Sea Lions and Marine Mammals in the Western Bering Sea." Kamchatka Branch, Pacific Institute of Geography, Russian Academy of Sciences.

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———. 2024. "Western Bering Sea Pollock — 2nd Surveillance Audit Report (SA2)." UCSL United Certification Systems Limited.